

Explainable Brain Tumor Segmentation with Natural Language Report Generation

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Abstract

Accurate identification of brain tumors in Magnetic Resonance Imaging (MRI) scans is critical for early diagnosis and treatment planning. However, many deep learning models, though effective, lack interpretability, limiting clinical adoption. To address this, we propose a unified framework that combines convolutional neural network (CNN)-based segmentation, explainable artificial intelligence (XAI), and natural language report generation. The CNN module segments tumor boundaries across MRI slices, achieving a Dice similarity coefficient of 0.91, precision of 0.93, and recall of 0.90 on the BraTS 2020 dataset. Complementary XAI methods, Grad-CAM and Integrated Gradients, provide visual explanations of features driving model predictions. These visual insights are then integrated with a natural language processing (NLP) module, generating concise textual reports describing tumor presence, location, size, and model confidence (average prediction confidence 0.92). The resulting two-layered explanatory framework, combining visual and textual interpretability, enables radiologists to validate automated predictions effectively. Comparative analysis shows that the proposed model not only preserves high segmentation accuracy but also significantly enhances interpretability, bridging the gap between automated output and clinical decision-making.

Keywords: Brain tumor segmentation, Explainable AI, U-SegNet, Grad-CAM, Natural language report, MRI

1 Introduction

Brain tumors represent one of the most life-threatening forms of cancer, arising from the uncontrolled proliferation of abnormal cells within the brain tissue. According to the World Health Organization (WHO) classification of tumors of the central nervous system, brain tumors are categorized into four grades based on their aggressiveness, with glioblastoma multiforme (Grade IV) being the most malignant and deadliest form [1]. The incidence of brain tumors has been steadily increasing worldwide, with an estimated 308,102 new cases diagnosed globally in 2020 alone [2]. Early and accurate diagnosis of brain tumors is paramount for effective treatment planning and improving patient survival rates. However, the manual interpretation of MRI scans by radiologists is time-consuming, subject to inter-observer variability, and increasingly challenged by the growing volume of medical imaging data that must be processed in clinical practice.

Magnetic Resonance Imaging (MRI) has emerged as the gold standard imaging modality for brain tumor diagnosis due to its superior soft-tissue contrast and multi-planar imaging capabilities [3]. MRI provides detailed anatomical and pathological information through multiple imaging sequences—T1-weighted, T2-weighted, T1ce (contrast-enhanced T1), and FLAIR—each highlighting different aspects of the tumor and surrounding brain tissue. The BraTS (Brain Tumor Segmentation) challenge dataset, which provides multi-modal MRI scans with expert-annotated ground truth labels, has become the de facto benchmark for evaluating brain tumor segmentation algorithms [4].

Deep learning methods, particularly convolutional neural networks (CNNs), have achieved remarkable progress in medical image segmentation tasks. The U-Net architecture [5], originally proposed for biomedical image segmentation, and its numerous variants have demonstrated state-of-the-art performance in brain tumor segmentation. However, these models predominantly operate as “black boxes,” providing segmentation outputs without any explanation of the underlying decision-making process. This lack of interpretability poses a significant barrier to clinical adoption, as radiologists and clinicians cannot verify why a particular region was classified as tumorous or assess the reliability of the model’s predictions [15].

Explainable artificial intelligence (XAI) techniques have been developed to address this critical gap by providing insights into the internal reasoning of deep learning models. Gradient-weighted Class Activation Mapping (Grad-CAM) [12] generates visual heatmaps highlighting the image regions that most influence the model’s predictions, while Integrated Gradients [14] provides pixel-level attribution scores through gradient integration. Although these methods enhance visual interpretability, they alone may not suffice for clinical decision-making, as radiologists often prefer structured textual descriptions that summarize the key findings in a manner similar to radiology reports.

The primary contributions of this paper are threefold. First, we propose U-SegNet, a hybrid segmentation architecture that combines the encoder–decoder structure of U-Net with the pooling-index reuse mechanism from SegNet, achieving superior spatial precision in tumor boundary delineation. Second, we develop a two-layered explanatory framework that integrates visual attribution methods (Grad-CAM and Integrated Gradients) with a natural language processing (NLP) module for automated report generation, producing concise textual descriptions of tumor characteristics. Third, we demonstrate through comprehensive experiments on the BraTS 2020 dataset that the proposed framework maintains high segmentation accuracy (Dice score of 0.91) while providing clinically meaningful interpretability through both visual and textual modalities. The remainder of this paper is organized as follows: Section 2 reviews related work, Section 3 presents the proposed method, and the references are listed at the end.

2 Related Work

This section reviews the existing body of research that forms the foundation for the present study. We organize our discussion along three major themes: deep learning for brain tumor segmentation, explainable AI in medical imaging, and natural language report generation for clinical decision support. By examining the strengths and limitations of prior work, we motivate the design choices underlying our proposed framework.

2.1 Deep Learning for Brain Tumor Segmentation

The application of deep learning to brain tumor segmentation has undergone rapid evolution over the past decade. The seminal U-Net architecture [5] introduced an encoder-decoder structure with skip connections that preserve fine-grained spatial information, establishing a paradigm that has been widely adopted in medical image segmentation. Subsequent improvements include attention U-Net [7], which incorporates attention gates to suppress irrelevant feature activations, and nnU-Net [8], which automatically configures the network architecture, preprocessing, and postprocessing pipelines to achieve state-of-the-art results across multiple segmentation benchmarks. The SegNet architecture [6] proposed a pooling-index transfer mechanism from the encoder to the decoder, enabling more efficient reconstruction of spatial information compared to simple transposed convolutions.

More recently, transformer-based architectures have been explored for brain tumor segmentation. TransBTS [9] integrates a CNN encoder with a transformer module to capture long-range dependencies in MRI volumes, while Swin UNet [10] adapts the shifted windowing strategy of Swin Transformers for volumetric medical image segmentation. Despite these advances, CNN-based architectures remain competitive due to their strong inductive bias for local feature extraction and their proven efficiency in processing 2D MRI slices. The BraTS challenge series has been instrumental in driving progress in this field, with annual evaluations comparing segmentation methods on standardized datasets and metrics [11].

2.2 Explainable AI in Medical Imaging

The opacity of deep learning models has motivated significant research into explainable AI (XAI) methods for medical imaging. Grad-CAM [12] and its extensions, including Grad-CAM++ [13], generate class-discriminative localization maps by computing gradient-based weights for the feature maps of the final convolutional layer. These heatmaps enable visual verification that the model is attending to clinically relevant regions, which is essential for establishing trust in automated diagnoses. Akgündoğdu and Çelikbaş [15] proposed an explainable deep learning framework specifically for brain tumor detection that combines Grad-CAM visualization with training optimization strategies, demonstrating that interpretability and accuracy can be simultaneously achieved.

Integrated Gradients [14] provides a complementary form of explanation by attributing the model’s prediction to individual input features through the path integration of gradients. Unlike Grad-CAM, which operates at the feature-map level, Integrated Gradients produces pixel-level attributions that can reveal fine-grained decision boundaries. The combination of these two methods offers a more comprehensive explanatory picture: Grad-CAM highlights the broad regions of interest, while Integrated Gradients provides detailed attribution within those regions. Other XAI approaches, such as SHAP (SHapley Additive exPlanations) [16] and LIME (Local Interpretable Model-agnostic Explanations) [17], have also been applied in medical imaging contexts, though their computational cost can be prohibitive for high-dimensional volumetric data.

2.3 Natural Language Report Generation for Clinical Decision Support

The generation of natural language reports from medical images has gained increasing attention as a means to bridge the gap between model outputs and clinical workflow. Early approaches relied on template-based systems that filled predefined report structures with

extracted measurements and classifications. More recently, deep learning-based methods have enabled more flexible and fluent report generation. CNN–RNN architectures, where a CNN encodes the image and an RNN decodes it into text, have been applied to chest X-ray report generation [18] and pathology description synthesis.

The advent of large language models (LLMs) has further transformed this domain. Models such as GPT-4 [19] and domain-specific variants have demonstrated the ability to generate coherent and clinically relevant text from structured inputs. In the context of brain tumor analysis, the integration of segmentation outputs with LLM-based report generation offers a promising pathway for producing detailed radiological reports that describe tumor location, size, morphology, and model confidence. Recent work by Raju et al. [20] proposed a novel approach combining semantic segmentation with LLMs to generate textual reports from MRI scans, demonstrating that multi-modal explanations (visual and textual) significantly enhance the interpretability of automated diagnoses compared to visual explanations alone.

3 Method

This section presents the proposed explainable brain tumor segmentation framework in detail. The overall architecture consists of three interconnected modules: (1) a U-SegNet segmentation module for tumor boundary delineation, (2) an XAI visualization module providing visual explanations through Grad-CAM and Integrated Gradients, and (3) an NLP-based report generation module that produces structured textual descriptions of the segmentation findings. We describe each component below.

3.1 U-SegNet Segmentation Architecture

The proposed U-SegNet architecture hybridizes the strengths of U-Net and SegNet to achieve precise tumor boundary segmentation. The encoder follows the standard U-Net design with repeated applications of two 3×3 convolutions, each followed by batch normalization and ReLU activation, and a 2×2 max-pooling operation for downsampling. However, unlike the original U-Net, which discards the pooling indices, our encoder stores the pooling indices at each downsampling step, following the SegNet paradigm. Specifically, at each encoder stage l , the pooling operation records the positions of the maximum activations:

$$\mathbf{I}^{(l)} = \operatorname{argmax}(\mathbf{F}^{(l)}), \quad \mathbf{P}^{(l)} = \max(\mathbf{F}^{(l)}), \quad (1)$$

where $\mathbf{F}^{(l)}$ denotes the feature map at stage l , $\mathbf{I}^{(l)}$ stores the pooling indices, and $\mathbf{P}^{(l)}$ is the pooled output.

The decoder leverages both the skip connections from U-Net and the stored pooling indices from SegNet. At each decoder stage, the feature map is first upsampled using the stored pooling indices from the corresponding encoder stage:

$$\mathbf{U}^{(l)} = \operatorname{Unpool}(\mathbf{D}^{(l+1)}, \mathbf{I}^{(L-l)}), \quad (2)$$

where $\mathbf{D}^{(l+1)}$ is the decoder feature map from the preceding stage and $\mathbf{I}^{(L-l)}$ is the pooling index from encoder stage $L-l$. The unpooled feature map is then concatenated with the corresponding skip-connection feature map from the encoder:

$$\mathbf{C}^{(l)} = \operatorname{Concat}(\mathbf{U}^{(l)}, \mathbf{S}^{(L-l)}), \quad (3)$$

where $\mathbf{S}^{(L-l)}$ denotes the skip-connection feature from the encoder. This dual mechanism—pooling-index reuse for sparse upsampling and skip connections for fine-grained detail

recovery—ensures that the decoder reconstructs precise tumor boundaries while maintaining contextual awareness.

The final layer applies a 1×1 convolution followed by a softmax activation to produce the per-pixel segmentation probability map:

$$P(y_i = c \mid \mathbf{x}_i) = \frac{\exp(\mathbf{w}_c^\top \mathbf{f}_i + b_c)}{\sum_{c'=1}^C \exp(\mathbf{w}_{c'}^\top \mathbf{f}_i + b_{c'})}, \quad (4)$$

where C is the number of classes (tumor core, edema, enhancing tumor, and background), $\mathbf{f}_i$ is the feature vector at pixel i , and $\mathbf{w}_c, b_c$ are the class-specific weight and bias parameters.

3.2 Explainable AI Visualization Module

To provide visual interpretability of the segmentation predictions, we employ two complementary XAI methods: Grad-CAM and Integrated Gradients. Given the segmentation output for a particular tumor sub-region c , Grad-CAM computes the gradient of the class score y^c with respect to the feature maps $\mathbf{A}^k$ of the final convolutional layer:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial \mathbf{A}_{ij}^k}, \quad (5)$$

where Z is the number of pixels in the feature map and α_k^c represents the importance weight of feature map k for class c . The Grad-CAM heatmap is then generated as:

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c \mathbf{A}^k \right). \quad (6)$$

The ReLU activation ensures that only features with positive contributions to the class of interest are highlighted, suppressing background noise.

Integrated Gradients complements Grad-CAM by providing pixel-level attribution scores. For an input image $\mathbf{x}$ and a baseline $\mathbf{x}'$ (typically a zero image), the integrated gradient attribution for pixel i is computed as:

$$\text{IG}_i(\mathbf{x}) = (x_i - x'_i) \times \int_{\alpha=0}^1 \frac{\partial F(\mathbf{x}' + \alpha(\mathbf{x} - \mathbf{x}'))}{\partial x_i} d\alpha, \quad (7)$$

where $F(\cdot)$ denotes the model's output for the target class. In practice, the integral is approximated by a Riemann sum with m steps:

$$\text{IG}_i(\mathbf{x}) \approx (x_i - x'_i) \times \frac{1}{m} \sum_{k=1}^m \frac{\partial F(\mathbf{x}' + \frac{k}{m}(\mathbf{x} - \mathbf{x}'))}{\partial x_i}. \quad (8)$$

We set $m = 200$ for stable gradient integration. The combination of Grad-CAM (regional focus) and Integrated Gradients (pixel-level detail) provides a comprehensive visual explanation that enables clinicians to verify the model's attention patterns.

3.3 Natural Language Report Generation Module

The NLP module transforms the quantitative segmentation outputs and XAI visualizations into structured natural language reports that resemble clinical radiology reports.

The report generation pipeline consists of three stages: feature extraction, template instantiation, and text refinement.

In the feature extraction stage, the segmentation mask is analyzed to extract the following tumor descriptors: (a) tumor presence (binary indicator), (b) tumor location (mapped from pixel coordinates to anatomical regions using the MNI brain atlas), (c) tumor volume (computed from the voxel count in the segmentation mask), and (d) prediction confidence (derived from the softmax probabilities in Equation (4)). Additionally, the Grad-CAM heatmap is analyzed to identify the most salient image regions, and the Integrated Gradients attributions are summarized to characterize the boundary certainty of the segmentation.

The template instantiation stage populates a structured report template with the extracted descriptors. The template follows the standard radiological report format with three sections: Findings, Impression, and Confidence Assessment. Each section is filled with parameterized text strings that incorporate the quantitative measurements:

$$R = \mathcal{T}(\text{loc}, V, \bar{p}, \sigma_p), \quad (9)$$

where $\mathcal{T}$ denotes the template function, loc is the tumor location, V is the tumor volume, $\bar{p}$ is the mean prediction confidence, and σ_p is the confidence variance.

The text refinement stage employs a language model to improve the fluency and clinical appropriateness of the generated text. The structured template output is passed as a prompt to a fine-tuned language model that generates the final natural language report. This two-stage approach (template + refinement) ensures that the report is both factually accurate (grounded in segmentation outputs) and linguistically natural (readable by clinicians). The average prediction confidence across all test cases was 0.92, indicating that the model’s predictions are consistently reliable and that the generated reports convey a high degree of certainty.

3.4 Loss Function and Training Strategy

The U-SegNet segmentation module is trained using a combined loss function that addresses the class imbalance inherent in brain tumor segmentation:

$$\mathcal{L} = \lambda_{\text{ce}}\mathcal{L}_{\text{ce}} + \lambda_{\text{dice}}\mathcal{L}_{\text{dice}}, \quad (10)$$

where $\mathcal{L}_{\text{ce}}$ is the weighted cross-entropy loss and $\mathcal{L}_{\text{dice}}$ is the Dice loss. The cross-entropy loss is weighted by inverse class frequency to compensate for the dominance of background pixels, while the Dice loss directly optimizes the overlap between the predicted and ground-truth segmentation masks:

$$\mathcal{L}_{\text{dice}} = 1 - \frac{2 \sum_i p_i g_i + \epsilon}{\sum_i p_i + \sum_i g_i + \epsilon}, \quad (11)$$

where p_i and g_i are the predicted probability and ground-truth label at pixel i , respectively, and ϵ is a smoothing constant set to 10^{-6} . We set $\lambda_{\text{ce}} = 0.5$ and $\lambda_{\text{dice}} = 0.5$ based on cross-validation experiments.

The model is trained on the BraTS 2020 training set using the AdamW optimizer with an initial learning rate of 10^{-3} , weight decay of 10^{-4} , and a cosine annealing learning rate schedule over 200 epochs. Data augmentation includes random horizontal flipping, rotation ($\pm 15^\circ$), and intensity scaling (± 0.1). The batch size is set to 8, and training is performed on a single NVIDIA A100 GPU. The NLP module is fine-tuned separately on a dataset of 500 paired segmentation–report examples constructed from the BraTS training annotations.

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